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Engineer building systems from FPGA firmware to production backends. Eight years of ownership over the core of a regional-scale product: billing for a million users, vehicle telemetry, route optimisation, geospatial analytics. Parallel track of personal projects in SDR, Bayesian price intelligence, and metacircular computation in Common Lisp. MIPT MSc, amateur radio licence R2BPW.

Own Projects

Radio Bargain Bot

Price-intelligence system for the secondary ham radio market. Aggregates listings from 98 sources across 50+ countries, scores each deal using Bayesian bootstrap benchmarks, and publishes verified bargains to 12 regional Telegram channels. Autonomous source discovery via Active Inference. 18+ months in continuous production. Peer-reviewed paper.

[r2bpw.works/radio_bargain_bot_v2](#) · [GitHub](#)

More Fun to Receive — Custom SDR Receiver on FPGA

Digital receiver written from scratch in Verilog for Xilinx Spartan-6 (OSA103 Mini): NCO, hardware-multiplier mixer, decimating CIC filter, UART interface. 200 MSPS. Board pinout reverse-engineered without documentation. PLL timing and FPGA startup sequence (COR2 register) debugged and fixed.

Development System — Self-Reproducing Computation in Common Lisp

Master/Apprentice architecture on SBCL with hot-loading of computational threads into a live Lisp image. Threads spawned via LLMs (OpenRouter), compiled and launched in the Apprentice. Metacircular: threads spawn threads that spawn threads. SICP-style minimalism — recursion, defun + let + cond + funcall, conditions & restarts. 93% Common Lisp.

[github.com/R2BPW/development_system](#)

IR Bridge — Ericsson MC218 ↔ Flipper Zero ↔ LLM

Bidirectional infrared bridge from a 1999 palmtop to a large language model. Native OPX drives the IrDA modulator at a baud rate producing a 38 kHz carrier; Flipper Zero decodes and relays via ESP32 to an LLM API. No cables, no IrDA stack.

[r2bpw.works/ir_bridge](#) · [GitHub](#)

Professional Experience

Backend Team Lead — Bolshaya Troika, Moscow

May 2018 – present · 8+ years

Core product for waste management logistics at regional scale (Moscow Oblast). Government contracts, international B2B clients.

Billing engine for 1M+ users — fast synchronous Python, strict typing, full test coverage. **Route optimisation** (TSP heuristics: nearest-neighbour, 2-opt, Or-opt) in pure Python, no external dependencies — 120 points in tens of milliseconds, haversine distance. Deployed as fallback optimiser.

Document processing with programmable semantics and versioning — used by international B2B clients. **Telemetry visualisation** and analysis across the entire regional vehicle fleet. Frontend and backend **internationalisation** (i18n).

Refactored codebase shared by 40+ developers. Introduced code review, raised test coverage. Responsible for releases including night shifts and rollbacks. Chaired the technical certification committee — signed product certification documents.

Teaching

Teaching assistant, MIPT (2022). Seminar instructor and code reviewer, Yandex.Practicum (2023–2024).

Education

MSc — Moscow Institute of Physics and Technology (MIPT), 2022. Department of Applied Mathematics and Informatics. Computer Science.

BSc — MIREA – Russian Technological University, 2020. Software Engineering.

Coursera: Practical Reinforcement Learning (HSE, 2021), Big Data Essentials: HDFS, MapReduce and Spark RDD (Yandex, 2020). Cisco CCNA (2019).

Technical Profile

Backend & data: Python, Django, DRF, Celery, PostgreSQL, ClickHouse, Hadoop/MapReduce, REST API, Pytest. High-load systems, geospatial data, real-time telemetry.

Algorithms & mathematics: Combinatorial optimisation (TSP, VRP, local search heuristics), Bayesian methods, statistical analysis, Active Inference. Haversine, bootstrap estimation.

Hardware & signals: FPGA (Xilinx Spartan-6), Verilog, DSP (DDC, NCO, CIC filters), SDR, antenna design and construction, amateur radio practice.

Systems & languages: Common Lisp (SBCL), C, Linux, Docker, CI/CD, Git. Metaprogramming, conditions & restarts. Two years on Gentoo (2013–2015). Wrote an ext2-type filesystem as a Linux kernel module.

Languages: Russian (native), English (B2).

Looking For

An environment where engineering depth is valued over velocity. Product development in B2C or an international context — or research and education where my profile (from FPGA to production) creates real value. Remote, Europe, or relocation.